IMPERIAL COLLEGE LONDON

DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING
EXAMINATIONS

MSc and EEE/EIE PART III/IV: MEng, BEng and ACGI

COMMUNICATION NETWORKS

Friday, 12 December 2025, 10:00 AM

Time allowed: 3:00 hours

There are FOUR questions on this paper.

Answer ALL questions.

All questions carry equal marks.

Any special instructions for invigilators and information for candidates are on page 1.

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Second Marker(s) : C. Ling

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Information for students

1. Mean delay for the M/M/1 system may be taken as

$$t_i = \frac{1}{\mu C_i - \lambda_i}$$

where,
$1/\mu$ = Average length of packet [bit/packet]
C_i = Transmission speed link i [bits/s]
μC_i = Service rate (link i) [packet/ s]
λ_i = Arrival rate (link i) [packet/ s]

2. Optimal Routing Problem (ORP)

$Min\ D(F)$ with respect to $F = \{F_i\}$

where, $D(F) = \sum_{i=1}^{L} \frac{F_i}{C_i - F_i}$

and,

C_i = Capacity of link l_i.
F_i = Flow carried by link l_i.

3. Velocity of propagation and propagation delay.

Velocity of unguided transmission through air or space of an electromagnetic wave is $\mathbf{2.99x10^8}$ m/s

Assume velocity of guided transmission to be approximately $\mathbf{66\%}$ of the unguided wave propagation.

And propagation delay:

$$\boldsymbol{a = \frac{Rd}{LV}}$$

Where,
$\boldsymbol{R}$ = Data rate of the channel.
$\boldsymbol{d}$ = Length of the network.
$\boldsymbol{L}$ = Average frame length.
$\boldsymbol{V}$ = Velocity of guided media.

The Questions

Q1.

a) Link Layer Protocols.

i) Briefly describe the Stop and Wait protocol using a half-duplex point to point communication link. **[2]**

ii) Derive a simple expression for the Utilisation of the Stop and Wait protocol. Clearly and explicitly define all the parameters and variables included in the derivation. **[2]**

b) Link Layer Protocols modified.

i) Propose a modified protocol to increase the Utilisation of the Stop and Wait protocol. Clearly define the protocol's parameters and the underlying physical characteristics of the network. **[2]**

ii) For the protocol, suggested in i): derive the Utilisation considering the underlying physical characteristics of the link and the protocol's parameters. **[2]**

iii) For the protocol, suggested in i): discuss how would you modify the expression of Utilisation if you need to consider retransmission of corrupted frames? **[2]**

c) Multimedia Networking.

In the context of supporting different classes of traffic on a packet network:

i) In your own words,

- discuss the potential benefits of scheduling incoming packets. **[2]**
- discuss and justify management of a server's input buffer (queue). **[2]**

ii) Using as a reference a standard M/M/1 queuing system, identify additional elements that will need to be in place to implement traffic shaping. **[3]**

iii) With the help of a sketch describe the operations of a Weighted Fair Queueing mechanism. **[3]**

Q2.

a) Transport Layer Protocols

i) Describe the operation of the following TCP congestion control mechanisms: I) additive increase / multiplicative decrease and II) reaction to time out events. **[3]**

ii) For the standard TCP protocol compare and contrast the ***slow start*** and ***congestion avoidance*** states. **[3]**

iii) Using an initial reference window size and knowledge of the network's round-trip time, discuss the advantages and disadvantages of UDP vs TCP connections? **[4]**

b) Mean Network Delay

For the packet network in **Fig. 2.1.** all external arrivals are assumed Poisson streams with mean arrival rate γ_{ij} Kbits per second. The traffic matrix T is given:

$$T = \begin{bmatrix} 0 & \gamma_{12} & \gamma_{13} & \gamma_{14} \\ \gamma_{21} & 0 & \gamma_{23} & \gamma_{24} \\ \gamma_{31} & \gamma_{32} & 0 & \gamma_{34} \\ \gamma_{41} & \gamma_{42} & \gamma_{43} & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0.15 & 0.10 & 0 \\ 0.36 & 0 & 0 & 0 \\ 0.25 & 0.20 & 0 & 0 \\ 0.15 & 0.15 & 0.15 & 0 \end{bmatrix}$$

All links have capacity: $C(i,j) = 4$ Kbits per second.

Packet lengths are exponentially distributed with mean length 6 Kbits/packet.

i) Derive the expression of mean network delay T. **[4]**

ii) You are permitted to upgrade the network adding one ***one-directional*** connection to alleviate the traffic in the bottleneck link. Identify and justify your chosen connection. **[4]**

iii) Derive the mean number of outstanding packets N' in the resulting network. Contrast the mean number of outstanding packets of the original network N with N'. **[2]**

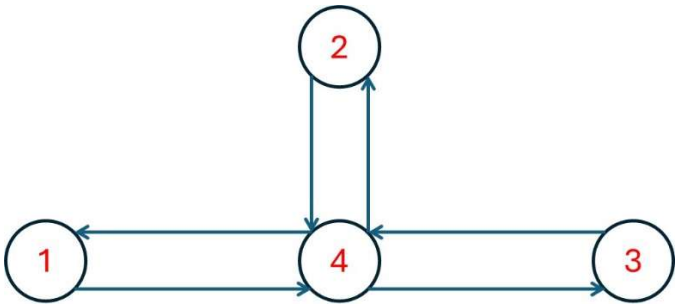


Figure 2.1.: Network.

Q3.

a) Wireless Networks Protocols

i) Describe two different wireless network scenarios where a standard collision detection mechanism will not have a significant impact in reducing collision of frames. **[3]**

ii) Using a time sequence diagram:

- Sketch, describe and discuss a collision avoidance (CA) mechanism known to you. **[2]**

- Sketch, describe and discuss a channel reservation mechanism (RM) known to you. **[3]**

iii) Using the time sequences sketched in ii): discuss the consideration to be made before selecting a CA versus a RM mechanism. **[2]**

b) Using the Network Utility Maximisation (NUM) objective function:

$$NUM_{obj} = \sum_{i} \frac{w(i)^{\alpha} r(i)^{1-\alpha}}{(1-\alpha)}$$

i) Describe and discuss what is meant by proportional fair flow assignment problem. **[2]**

ii) Solve the proportional fair flow assignment for the network shown in **Fig. 3.1**. **[3]**

iii) Describe and discuss the meaning of a weighted proportional fair allocation principle. **[2]**

iv) Solve the weighted proportional fair flow assignment problem for the network in **Fig.3.2**. **[3]**

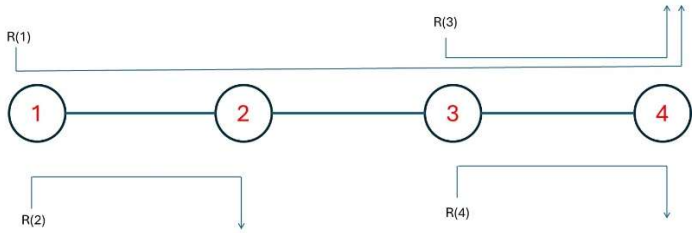


Figure 3.1: NUM flow allocation problem.

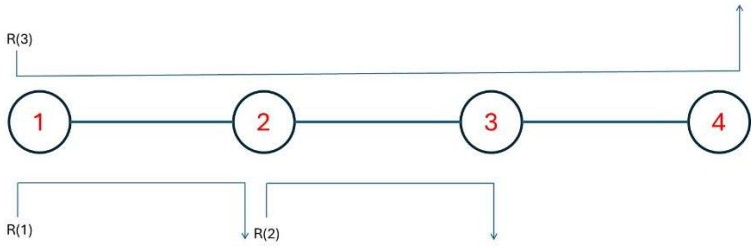


Figure 3.2: NUM flow allocation problem.

Q.4.

a) Optimal Routing Problem

i) Describe and discuss the differences between the *Routing Information* (RIP) routing protocol and the *Open Shortest Path First* (OSPF) routing protocol. **[4]**

ii) For the network in **Fig. 4.1** and the given costs of using $Path_1$, $Path_2$, $Path_3$:

$$\text{Cost Path_1} = \frac{1}{3}\text{x}(1)^3$$
$$\text{Cost Path_2} = \text{x}(2)^2$$
$$\text{Cost Path_3} = \text{x}(3)$$

- Solve the Optimal Routing Problem (ORP). **[4]**

- Discuss your findings. **[2]**

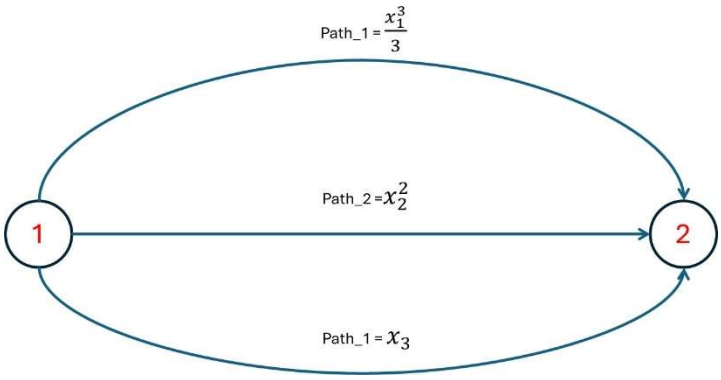


Figure 4.1.: Network.

b) Low Altitude Vehicular (LAV) Networks.

In the context of a low-altitude network of vehicles:

i) Describe and discuss potential applications of LAV networks. **[3]**

ii) Identify the key aspects that affect the operations of LAV networks. **[3]**

iii) Discuss in your own words how satellite and terrestrial networks can support LAV networks. **[4]**